

1. Administrative & Study Identification

**Full Study Title** A Phase 3, Randomized, Double-Blind, Placebo-Controlled Program to Evaluate the Efficacy and Safety of MK-7240 in Participants with Moderately to Severely Active Ulcerative Colitis **Short Title/Acronym** ATLAS-UC **Protocol Number** MK-7240-001 **NCT Number** NCT06052059 **Sponsor Name** Merck Sharp & Dohme LLC **Investigational Product** Tulisokibart (MK-7240) (anti-TL1A monoclonal antibody) **Phase of Development** Phase III **Medical Monitor** Merck Clinical Operations Lead

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2. Study Synopsis & Rationale

**2.1 Study Objective Primary Objective:** To evaluate whether at least one tulisokibart dose level is superior to placebo in the proportion of participants achieving clinical remission according to the Modified Mayo Score at Week 12 and Week 52. **Secondary Objectives:** To evaluate endoscopic improvement, clinical response, long-term safety, and the potential reduction of intestinal fibrosis associated with inflammatory bowel disease.

**2.2 Background & Rationale** To address the persistent therapeutic ceiling and high unmet medical need in patients with moderately to severely active ulcerative colitis. Tulisokibart is an investigational humanized monoclonal antibody directed to a novel target, tumor necrosis factor (TNF)-like cytokine 1A (TL1A). Clinical studies suggest that targeting TL1A may inhibit inflammatory pathways and help reduce intestinal fibrosis, which may be critical in altering disease progression and achieving long-term clinical remission.

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3. Study Design & Methodology

**3.1 Design Framework Study Type:** Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized

**3.2 Planned Enrollment & Duration Target** **\$N\$:** 1,020 participants **Number of Sites:** Global, multi-center **Estimated Study Start:** 25-Oct-2023 **Estimated Primary Completion:** 21-Nov-2026

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4. Subject Selection (Eligibility Criteria)

**4.1 Inclusion Criteria** 1. Age 16 to 80 years. 2. Confirmed diagnosis of ulcerative colitis (UC) for at least 3 months with

moderately to severely active disease. 3. Inadequate response, loss of response, or intolerance to conventional treatments (e.g., corticosteroids, immunosuppressants) or advanced therapies.

**4.2 Exclusion Criteria** 1. Diagnosis of Crohn's Disease (CD), indeterminate colitis, or UC limited to the rectum. 2. Prior or current evidence of definite colonic dysplasia or history of cancer within the last 5 years. 3. Current diagnosis of fulminant colitis, toxic megacolon, or impending need for colostomy/ileostomy.

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## 5. Intervention & Treatment Plan

**5.1 Investigational Regimen Dosage/Frequency:** Tulisokibart at different pre-specified dose levels or matching placebo administered across induction and maintenance phases. **Route of Administration:** Intravenous (IV) infusion during the induction period and Subcutaneous (SC) injection during the maintenance period. **Duration of Treatment:** Up to 52 weeks (1 year) for the main study, with a potential extension period of up to 3 years for patients demonstrating response.

**5.2 Concomitant & Prohibited Therapy Permitted:** Standard baseline UC medications (e.g., 5-ASAs) provided they remain at a stable dose. **Prohibited:** Concurrent investigational therapies, targeted advanced therapies for IBD outside of the protocol, or receipt of live virus vaccinations during the treatment period.

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## 6. Study Timeline & Visit Schedule

| Visit ID                 | Window (Days)  | Key Activities/Procedures                                            |
|--------------------------|----------------|----------------------------------------------------------------------|
| Screening                | Up to 6 weeks  | Informed Consent, Medical History, Endoscopy, Laboratory Tests       |
| Induction                | Weeks 0 to 12  | Randomization, IV Infusions (5 visits), Safety Assessments           |
| Reinduction (If needed)  | Up to 3 months | IV Infusions for patients not responding well to initial induction   |
| Maintenance              | Weeks 12 to 52 | SC Injections (7 visits), Efficacy Assessment, Modified Mayo Score   |
| End of Study / Follow-Up | Post Week 52   | Final Endoscopy, Exit Interview, Safety Follow-Up (approx. 14 weeks) |

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## 7. Outcome Measures (Endpoints)

**7.1 Primary Endpoint Definition:** Proportion of participants achieving clinical remission according to the Modified Mayo Score at Week 12 (Study 2) and both Week 12 and Week 52 (Study 1). **Measurement Method:** Clinical assessment combined with central reading of endoscopy for mucosal healing.

**7.2 Secondary & Exploratory Endpoints** \* Proportion of participants with Endoscopic Improvement. \* Proportion of participants achieving Clinical Response.

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## 8. Safety & Adverse Event Reporting

**Adverse Event (AE) Monitoring:** Continuous monitoring throughout the study via physical examinations, vital signs, laboratory safety tests, and patient-reported symptoms. **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness to the sponsor. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee (iDMC) to conduct unblinded safety reviews.

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## 9. Statistical Analysis Plan (SAP) Summary

**Analysis Populations:** Intent-to-Treat (ITT) for primary efficacy analyses. Safety Set for all participants who received at least one dose of the study drug. **Sample Size Justification:** Approximately  $N=1,020$  participants, sufficiently powered ( $\alpha = 0.05$ ,  $\text{Power} \geq 80\%$ ) to evaluate the superiority of at least one dose level of tulisokibart over placebo. **Handling of Missing Data:** Non-responder imputation (NRI) for binary efficacy endpoints and Multiple Imputation for missing continuous data.

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## 10. Quality Assurance & Regulatory Compliance

**GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular on-site and remote monitoring visits to assure protocol adherence and data integrity. **Audit Trail:** Electronic Case Report Form (eCRF) utilizing 21 CFR Part 11 compliant systems, with full query management and source document verification.

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## 11. Study Identifiers & Governance

**Primary ID:** MK-7240-001 **Registry Number:** NCT06052059 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC **Study Title:** ATLAS-UC **Official Regulatory Title:** A Phase 3, Randomized, Double-Blind, Placebo-Controlled Program to Evaluate the Efficacy and Safety of MK-7240 in Participants with Moderately to Severely Active Ulcerative Colitis

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## 12. Clinical Framework (Ulcerative Colitis Specific)

**Primary Condition:** Moderately to Severely Active Ulcerative Colitis **Diagnosis Threshold:** Modified Mayo Score consistent with

moderate to severe disease activity. **Medical Methodology:** Targeted biologic therapy (Anti-TL1A monoclonal antibody). **Optimization Target:** Clinical Remission and Endoscopic Improvement at Weeks 12 and 52.

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### 13. Study Procedures & Methodology

**Management Pathway:** Step-up biological therapy for moderate-to-severe UC refractory to conventional treatments. **Education Protocol (HCP):** Site initiation visits and standardized endoscopic scoring training for investigators. **Education Protocol (Patient):** Informed consent process detailing IV to SC transition and potential symptom monitoring. **Quality Audit Mechanism:** Centralized reading of endoscopies/biopsies to ensure standardized Mayo subscore grading.

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### 14. Observation & Follow-Up Schedule

**Total Duration:** Up to 52 weeks (plus up to 3 years optional extension) **Onsite Visit Cadence:** Frequent visits during Induction (5 visits) and Maintenance (7 visits). **Remote Monitoring Frequency:** Continuous safety and diary entries via ePRO (electronic Patient-Reported Outcomes). **Checklist Review Interval:** Routine reviews prior to each administration of the study drug.

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### 15. Sign-off & Approval

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|--------------------------|--------|-----------|---------------|-----|------|-----|-----|
| Role                     | Name   | Signature | Date          | :-- | :--  | :-- | :-- |
| Principal Investigator   | [Name] | _____     | [DD-MMM-YYYY] |     |      |     |     |
| Clinical Operations Lead | [Name] | _____     | [DD-MMM-YYYY] |     | Lead |     |     |
| Statistician             | [Name] | _____     | [DD-MMM-YYYY] |     |      |     |     |